

SafeSite AI: Automated Construction-Site Safety Monitoring

Using Object Detection and Rule-Based Spatial Reasoning

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Abstract—Enforcement of personal protective equipment (PPE) rules on construction sites still depends largely on manual CCTV supervision, which does not scale to continuous monitoring of every worker. This paper presents SafeSite AI, which detects workers, helmets, vests and machinery, determines per-worker compliance through spatial reasoning, and produces an interpretable risk score. A unified six-class dataset of 10,680 images was built by standardising five public sources with incompatible labelling conventions; pixel-level hashing removed 213 duplicate images, 202 of them shared between different source datasets. Five detectors were trained under identical conditions and evaluated on 1,069 held-out images. YOLOv8s reached 0.7258 mAP@50 at 88.1 FPS, while YOLOv10n reached 165.0 FPS at 0.6676. Because no public dataset annotates which helmet belongs to which worker, compliance is resolved geometrically rather than learned. Analysis identifies two separable causes of per-class weakness: 72.7% of No_Helmet annotations fall below the COCO small-object threshold, while No_Vest is limited by a 50:1 class imbalance that persists uniformly across all five architectures.

Index Terms—construction safety, object detection, YOLO, personal protective equipment, dataset standardisation

I. INTRODUCTION

A. Problem Statement

Construction ranks among the most hazardous industrial sectors. Workers share space with heavy machinery, work at height, and pass beneath zones where objects fall. Protection rests on two mechanisms: correct use of protective equipment, and exclusion from restricted areas.

Both are currently enforced by human observation, which fails in three structural ways. One observer cannot attend to many camera feeds at once, sustained visual vigilance degrades within hours, and a supervisor occupies one location at a time. Violations consequently pass unrecorded, producing no signal until an injury occurs.

This work therefore targets automatic detection of safety violations from site imagery, with output specific enough to act on: not that a violation exists somewhere in a scene, but which worker is non-compliant and in what respect.

B. Objectives

- 1) Build a validated PPE detection dataset by standardising multiple public sources with differing class vocabularies and annotation formats.
- 2) Compare five detection architectures under identical conditions across accuracy, inference speed and model size.
- 3) Associate detected equipment with individual workers to yield per-worker compliance decisions.
- 4) Aggregate violations into a documented site-level risk score.
- 5) Quantify the resulting limitations, distinguishing those caused by data from those caused by model choice.

C. Literature Review

Two detector families dominate the field: two-stage architectures such as Faster R-CNN [1], favouring accuracy, and one-stage architectures such as YOLO, favouring inference speed. Construction sites stress both, since workers, machinery, equipment and hazardous zones must be tracked concurrently.

Surveying 66 studies, Arshad et al. [2] position computer vision and IoT as the two leading approaches to construction safety management, favouring vision for its non-intrusive use of existing cameras, and naming object tracking and depth perception as the outstanding research directions. Barlybayev et al. [3] benchmarked YOLOv8 variants on datasets including CHV and SHEL5K — both used here — and found larger variants more accurate but proportionately more expensive to train and run.

Deployment conditions complicate matters further. Wang et al. [4] targeted tunnel construction, where dim lighting and distant workers degrade recognition; their modified YOLOX with ConvNeXt blocks and CLAHE enhancement reached 86.94% mAP at roughly 22 FPS. Liu et al. [5] reported parallel difficulties from occlusion and viewing angle, raising helmet-detection mAP from 88.3% to 92.3% using FasterNet, Wise-IoU and CBAM attention while simultaneously shrinking the

model. Both results indicate that small equipment and adverse conditions, not architecture alone, govern practical accuracy. Liu et al. [6] argue more broadly that single-task detection yields only partial situational understanding, and combine detection, segmentation, pose estimation and tracking within one framework — a position that directly motivates treating detection here as input to reasoning rather than as output.

Three datasets underpin the present work. Otgonbold et al. [7] rebuilt the SHD helmet dataset as SHEL5K, completing its missing labels and expanding three classes to six across 5,000 images. Benchmarking ten detectors, YOLOR led at 88.28% mAP@0.5 with 0.012 s inference while Faster R-CNN with Inception V2 trailed at 36.89%. Its 416×416 images hinder small-object detection, and it never links equipment to a specific worker. Wang et al. [8] screened close to 10,000 images to retain 1,330 for the CHV dataset, spanning workers, vests and four helmet colours; YOLOv5x led on accuracy at 86.55% mAP and YOLOv5s on speed at 52 FPS, and privacy-motivated face blurring cost roughly 7% helmet accuracy. Containing only still frames, it cannot support tracking or temporal analysis.

Two further works address gaps this project inherits. Wang [9] observes that enforced regulation makes genuine violation imagery scarce, and proposes a domain-adaptive Faster R-CNN drawing on general-context images alongside construction data; a ResNet-152 backbone reached 86.84% mAP on unseen sites, exceeding supervised baselines by up to 14 points, though at a computational cost that questions real-time use. Yang et al. [10] note that workers may carry equipment without wearing it, and detect PPE jointly with three body keypoints, applying geometric rules to judge correct wear; their framework reached 92.9% mAP@0.5 at 155 FPS on 13.42 M parameters, and their ablation shows the rule layer contributing more than the accompanying attention module.

Four themes recur. Detection degrades on small, distant or occluded equipment. Adverse illumination increases misses. Class imbalance yields strong frequent classes and weak rare ones. And detecting an object does not establish whether a situation is safe — the spatial relationships between workers, equipment and hazards must also be resolved. Accordingly, this project standardises a multi-source dataset, compares five architectures under controlled conditions, and resolves compliance per worker through spatial association rather than by flagging a violation class anywhere in frame. The claimed contribution is not PPE detection itself, but this combination within one prototype together with an evidenced account of which limitations arise from data and which from models.

D. Dataset Identified

TABLE I
SOURCE DATASETS

Dataset	Format	Images	Contribution
SHEL5K [7]	VOC XML	5,000	Helmet, person
SHWD [11]	VOC XML	7,581	Helmet / no-helmet
CHV [8]	YOLO	1,330	Vest, person, helmet
Roboflow CSS v30 [12]	YOLO	717	Machinery, vest
Ultralytics PPE [13]	YOLO	1,416	Supplementary PPE

SHEL5K, SHWD and CHV are distributed for research use; the Roboflow dataset is CC BY 4.0 and the Ultralytics dataset AGPL-3.0. The Roboflow set was pinned to version 30 (workspace roboflow-universe-projects, project construction-site-safety) since community datasets on that platform are periodically revised.

II. METHODOLOGY

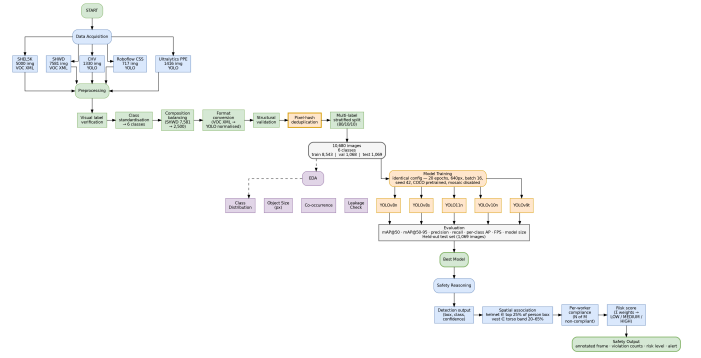


Fig. 1. System workflow. Five sources are standardised into one dataset; five detectors are trained under identical conditions; the best feeds a rule-based reasoning layer producing per-worker compliance and a risk score.

A. Label Semantic Verification

The five sources employ mutually incompatible vocabularies, and identical terms do not always denote identical objects. Every class of every source was therefore verified visually: annotated regions were cropped and displayed in labelled grids for inspection.

This exposed a conflict that would otherwise have propagated undetected. SHWD’s person label denotes not a human body but a *bare head* — a head lacking a helmet. Mapping it onto a unified Person class would have taught the detector that a person is a small head-sized region, and since test annotations would encode the identical error, no metric could have revealed it. The label was instead mapped to No_Helmet.

B. Class Standardisation and Balancing

A twelve-class intermediate vocabulary was defined via per-source mapping dictionaries, then narrowed to six: Person, Helmet, No_Helmet, Vest, No_Vest and Machinery. CHV’s four colour-specific helmet classes were

collapsed, since the safety rule concerns presence rather than colour and the distinction would have quartered examples per class. SHEL5K’s `person_with_helmet` and `person_no_helmet` both map to `Person`: compliance is deliberately excluded from the class vocabulary and resolved geometrically instead (Section II-G), so that additional equipment types become rule changes rather than retraining. Classes overlapping retained classes on identical pixels (`face`, `head_with_helmet`) were discarded as contradictory supervision.

An audit performed before any files were written showed SHWD alone contributing 111,514 `No_Helmet` annotations against roughly 34,000 helmets from all sources combined. SHWD was therefore subsampled from 7,581 to 2,500 images with a fixed seed, drawn 70% from helmet-containing images. The imbalance was diluted by controlling source proportion rather than removed at source.

C. Format Conversion and Validation

Pascal VOC annotations were converted to YOLO format. For corners $(x_{\min}, y_{\min})$ and $(x_{\max}, y_{\max})$ in an image of width W and height H :

$$x_c = \frac{(x_{\min} + x_{\max})/2}{W}, \quad w = \frac{x_{\max} - x_{\min}}{W} \quad (1)$$

with y_c and h defined analogously. Normalisation to $[0, 1]$ makes annotations invariant to resizing and renders range violations trivially detectable. Out-of-bounds coordinates were clamped and degenerate boxes discarded. Filenames were rewritten to embed source provenance, which later enables source-aware stratification. Structural validation confirmed complete image–label pairing, valid class indices and in-range coordinates, with no failures.

D. Deduplication

Duplicates were identified by hashing decoded RGB pixels rather than file bytes, since a photograph re-encoded at different compression produces different bytes but identical pixels. This found 212 duplicate groups covering 213 redundant images — 202 of them *cross-source*, the same photographs appearing in both CHV and SHWD under different filenames. An audit confirmed the copies carried complementary rather than conflicting annotations, so they were merged rather than deleted, with same-class boxes above IoU 0.90 treated as duplicates. The same-class restriction preserves legitimately nested annotations such as a helmet inside a person region.

Had these survived partitioning, roughly 40 images would have entered validation or test with identical copies in training, inflating every reported figure in a manner no metric could expose.

E. Partitioning

Since each image carries several class labels simultaneously, conventional stratification does not apply. Multi-label stratified shuffle splitting was used, each image represented by an eleven-dimensional binary vector encoding presence

of the six classes and the five sources. Including source membership gives every partition a representative mixture of contributing datasets. Splitting proceeded in two stages (80/20, then halving the remainder) under a fixed seed, yielding 8,543 training, 1,068 validation and 1,069 test images. Independent verification confirmed zero image overlap between any pair of partitions.

F. Model Training

Five architectures were trained: YOLOv8n, YOLOv8s, YOLO11n, YOLOv10n and YOLOv9t. Four are nano or tiny scale and differ principally in architectural generation, permitting comparison at approximately fixed capacity; YOLOv8s shares YOLOv8n’s architecture at roughly four times the parameters, isolating capacity alone. The design thus separates two variables commonly confounded: architectural progress and model size.

Conditions were identical throughout — same partitions, 20 epochs, 640×640 input, batch 16, seed 42, patience 5, COCO-pretrained initialisation. Mosaic augmentation was disabled for the full run via `close_mosaic`, following a preliminary run in which enabling it lowered accuracy across all five architectures; remaining augmentation comprised HSV jitter (0.015/0.7/0.4), horizontal flip ($p=0.5$), translation 0.1 and scaling 0.5. Training used a single NVIDIA Tesla T4. Evaluation used only the held-out test partition, which played no part in checkpoint selection.

G. Spatial Association and Risk Scoring

Detection localises workers and equipment independently but establishes no correspondence between them, and no source annotation encodes such a relation. Compliance is therefore geometric. For a person region spanning $[p_{y1}, p_{y2}]$, a helmet is associated with that worker when its centroid falls within the horizontal extent of the person box and the upper quarter of its vertical extent:

$$p_{y1} \leq h_c^y \leq p_{y1} + 0.25(p_{y2} - p_{y1}) \quad (2)$$

Restricting to the head region rather than the whole box prevents a helmet carried at waist height from registering as compliance. Vests use a torso band spanning 20–65% of the person region.

Violations aggregate into a risk score weighted 20 per missing helmet and 15 per missing vest, with 30 and 25 reserved for restricted-area entry and machinery proximity. Weights follow a safety argument rather than empirical calibration: head injury carries the highest fatality risk among preventable incidents, whereas vest absence principally affects visibility. Scores cap at 100 and map to Low (≤ 30), Medium (≤ 60) and High bands.

This layer is rule-based by design. No dataset supplies ground-truth risk labels against which such a model could be trained, and an explicit formula remains auditable where a learned score would not. The ablation of Yang et al. [10], where an equivalent rule layer outperformed their attention module, supports treating spatial rules as a contribution rather than a compromise.

III. RESULTS

A. Data Analysis



Fig. 2. Training-set class distribution. Helmet (23,859) exceeds No_Vest (475) by roughly 50:1, as only one source annotates vest violations.

The training partition holds 71,917 annotations: Helmet 23,859, No_Helmet 22,456, Person 21,709, Vest 3,028, No_Vest 475 and Machinery 390. The resulting 50:1 imbalance predicted the system’s principal weakness before any model was fitted.

TABLE II
OBJECT SCALE BY CLASS (TRAINING PARTITION)

Class	n	Small	Med.	Large	Median
Person	21,709	8.4%	39.2%	52.4%	100 px
Helmet	23,859	54.4%	34.4%	11.2%	29 px
No_Helmet	22,456	72.7%	21.9%	5.4%	23 px
Vest	3,028	6.1%	26.6%	67.3%	139 px
No_Vest	475	24.4%	33.7%	41.9%	73 px
Machinery	390	2.8%	12.3%	84.9%	259 px

Table II shows the two weakest classes failing for distinct reasons. Helmet classes are dominated by small objects: 72.7% of No_Helmet and 54.4% of Helmet annotations fall below the 32-pixel COCO threshold, at median equivalent sides of 23 and 29 pixels. Machinery is conversely 84.9% large at a median of 259 pixels, so its weakness cannot be attributed to scale and instead follows from having only 390 training instances.

Crowding analysis gave a mean of 8.4 objects and 2.5 persons per image (medians 5 and 1); 23.6% of images carry no Person annotation, consistent with SHWD supplying head-only labels, and 18.7% carry five or more persons.

Co-occurrence analysis validates the association layer directly. Among images containing a Vest annotation, 96% also contain a Person annotation; the figure is 81% for Helmet and 100% for No_Vest. The rule therefore has the co-occurring annotations it requires across the large majority of relevant images. Machinery co-occurs with Person in only 55% of cases, constraining validation of the proximity feature. Per-source median normalised box heights for No_Helmet ranged from 0.043 (SHWD) to 0.143 (CHV), all consistent with head-sized rather than body-sized regions, confirming uniform annotation convention across contributing sources.

B. Model Comparison

TABLE III
COMPARISON ON THE HELD-OUT TEST SET (1,069 IMAGES)

Model	mAP@50	mAP@50-95	P	R	Par.	FPS
YOLOv8s	0.7258	0.4594	0.7920	0.6783	11.13 M	88.1
YOLOv8n	0.7066	0.4402	0.8011	0.6546	3.01 M	109.5
YOLOv9t	0.7049	0.4466	0.7825	0.6630	1.97 M	100.6
YOLO11n	0.6921	0.4338	0.8232	0.6293	2.58 M	123.2
YOLOv10n	0.6676	0.4194	0.7750	0.6084	2.27 M	165.0

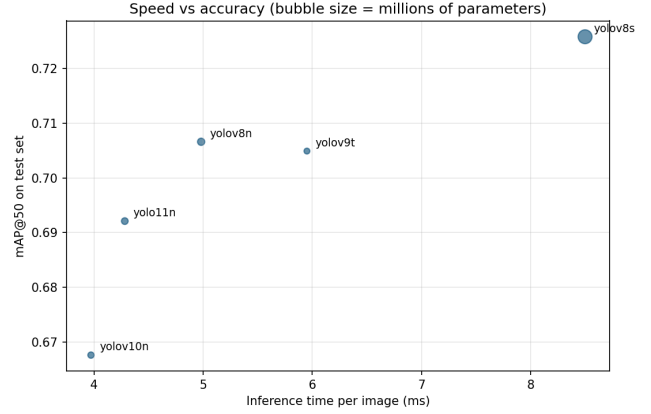


Fig. 3. Inference time against test accuracy; marker area is proportional to parameter count.

Two findings follow. The capacity ablation succeeded: YOLOv8s exceeds YOLOv8n by 1.9 mAP@50 points at roughly four times the parameters, while throughput falls from 109.5 to 88.1 FPS. Capacity therefore constrains this task partially, but the gain is a trade rather than a free improvement.

Second, the four similarly-sized models span only 3.9 mAP@50 points (0.6676–0.7066) while their speeds differ by 64% (100.6–165.0 FPS). Architectural generation thus governs computational cost far more strongly than accuracy here, implying data rather than architecture as the limiting factor. YOLOv10n illustrates this: fastest by a 34% margin, consistent with its NMS-free design removing the post-processing stage, yet least accurate. All five exceed the 25–30 FPS needed for live CCTV, the slowest by roughly threefold.

C. Per-Class Performance

TABLE IV
PER-CLASS MAP@50 ON THE TEST SET

Model	Person	Helm.	No_H.	Vest	No_V.	Mach.
YOLOv8s	0.8886	0.8660	0.8654	0.8098	0.2838	0.6411
YOLOv9t	0.8829	0.8545	0.8325	0.7711	0.2428	0.6454
YOLOv8n	0.8713	0.8551	0.8254	0.7798	0.2906	0.6172
YOLO11n	0.8703	0.8566	0.8325	0.7920	0.2290	0.5725
YOLOv10n	0.8535	0.8407	0.8166	0.7501	0.1977	0.5472

Aggregate figures are carried by the three frequent classes: for the best model Person, Helmet and No_Helmet all exceed

0.86 and Vest reaches 0.8098, against 0.2838 for No_Vest and 0.6411 for Machinery.

Critically, No_Vest performs poorly under *every* architecture, spanning 0.1977 to 0.2906. An architectural deficiency would be expected to leave at least one model substantially better; the uniformity across five distinct designs instead establishes data scarcity as the cause. Both No_Vest and Machinery carry roughly 50 test instances, so inter-model differences on those columns represent noise rather than architectural superiority.



Fig. 4. Normalised confusion matrix, YOLOv8s.

Fig. 4 shows negligible class-to-class confusion; the dominant error for every class is confusion with background. 84% of No_Vest and 44% of Machinery instances are predicted as background, indicating missed detections rather than misclassification, while 32% of background predictions are labelled Person and 31% Helmet, representing false positives on unannotated regions.

Validation mAP@50 rose from 0.506 at epoch 1 to 0.704 at epoch 20 without plateauing, and early stopping never triggered. The models are therefore undertrained rather than overfitted.

D. Safety Reasoning



Fig. 5. YOLOv8s detections on unseen test images.

Applying the rules of Section II-G to unseen test images produced per-worker verdicts and risk scores from 0/100 (Low) for a single compliant worker, through 35/100 (Medium) for a two-worker scene with one helmet and one vest violation, to 95/100 (High) for a five-worker scene lacking one helmet and five vests. Each output names the specific non-compliant worker rather than signalling that a violation exists somewhere in frame.

IV. DISCUSSION

A. Limitations

Vest violations. No_Vest reached 0.2838 mAP@50, traceable to 475 training instances against 23,859 helmets, since only one source annotates this class. Wang [9] characterises this as structural to the field, since enforced regulation makes genuine violation imagery scarce. A mitigation exists within the present design: deriving vest violations from the far stronger Vest class (0.8098) rather than direct No_Vest detection.

Small objects. With 72.7% of No_Helmet and 54.4% of Helmet annotations below the small-object threshold, detail is lost through downsampling at 640×640 ; raising input resolution is the principal remedy available, at proportionate cost.

Undertraining. Validation performance had not plateaued at 20 epochs. Training was bounded by available GPU allocation, so reported figures represent a lower bound.

One-stage detectors only. No two-stage baseline was trained, for compute reasons. Otgonbold et al. [7] report Faster R-CNN with Inception V2 at 36.89% mAP@50 against 88.28% for the best one-stage detector on a comparable dataset, at roughly four times the inference cost.

No external validation. All partitions derive from the same five sources, so test performance measures generalisation to further images of the same distribution rather than to an unseen

site. The 5,081 SHWD images excluded during balancing remain untouched by training and constitute a ready external test set.

Further constraints. Vests resemble ordinary reflective workwear, a confusion Yang et al. [10] report even at 93.3% vest mAP. All sources comprise still images, precluding fall detection and behaviour analysis, which are defined by motion. Risk weights follow reasoning rather than injury statistics and would require calibration against incident records. Public datasets depict predominantly European, North American and East Asian sites, leaving performance under Bangladeshi conditions unverified. Finally, continuous identified surveillance raises questions of consent and retention; Wang et al. [8] show privacy-preserving face blurring is feasible at roughly 7% helmet accuracy.

B. Future Work

Extending training beyond 20 epochs is the most immediate available improvement. Evaluating on the retained SHWD partition would supply an external generalisation measure at no data cost, and raising input resolution to 960 pixels directly targets the small-object limitation. Multi-object tracking would enable restricted-zone entry detection and suppress repeated alerts for a stationary violator. A locally annotated test set would quantify domain shift for Bangladeshi sites, a measurement absent from existing benchmarks.

V. CONCLUSION

SafeSite AI combines object detection with rule-based spatial reasoning for construction-site safety monitoring. A six-class dataset of 10,680 images was assembled from five sources with incompatible conventions, requiring visual verification of class semantics, controlled composition balancing, and pixel-level deduplication that removed 213 duplicates, 202 of them shared across different source datasets.

Five architectures trained under identical conditions were evaluated on a held-out partition. YOLOv8s led at 0.7258 mAP@50 while sustaining 88.1 FPS, and all five operate above real-time requirements. The capacity ablation showed a measurable but modest benefit from model size, while four similarly-sized architectures differed by under four mAP points, indicating data rather than architecture as the limiting factor.

Analysis isolated two independently evidenced causes of per-class weakness: small object scale for helmet classes, and a 50:1 imbalance for vest violations that persists uniformly across all five architectures and is therefore attributable to data rather than design. Extending the training schedule, followed by evaluation against the retained external partition, constitutes the most valuable next step.

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